Governing relations of the $CaCO_3$-barrier screen: derivation

Sessions P0-W3.1, P0-W3.2 (with the cp2 fix-up) and P0-W3.3 (cp3 amendments), contract v3 (symbolic; no values)

Each section opens with the physical argument, and the algebra follows it. Every displayed relation carries its SI units in brackets. The notation is that of `theory/symbols.yaml`, the functions are those of `theory/contract.yaml`, and open choices for the sign audit are listed in `theory/open_questions.md` (cited here as Q1, Q2, ...).

1 Regimes and what each link must achieve

Physics. The proposed mechanism is assembled in one chemical environment and must survive in another. The template is anchored, binds Ca^{2+} and templates the mineral while the electrode is a water-processed, alkaline slurry at room temperature (R-slurry). The finished interface then sits in a non-aqueous, fluorine-free electrolyte with the Al collector held at the cathode's anodic potential (R-cell). The two regimes have different solvents, different reference states and different corrosion modes, so a relation is meaningful only in the regime whose reference states it uses. Every relation below is therefore tagged with its regime, and none is carried across without saying so.

Table 1 states, for each link, the necessary condition the link must meet in its regime. These are directions of inequalities, not thresholds: the P0 thresholds, margins and decision rules are in `theory/thresholds.md`. Meeting them makes the mechanism thermodynamically possible under the stated framing; it does not show that a barrier works, and no rate, capacity or device statement follows from any of them.

Link	Regime	What the link must achieve	Relation
C1 anchor	R-slurry	template held at the oxide against water	$W_{\mathrm{adh,wet}} > 0$, Eq. (1)
C2 sequester	R-slurry	Ca^{2+} held at the template against hydrated Ca^{2+}	$\Delta G_{\mathrm{seq}} < 0$, Eq. (7); it acts on C3 only through $\Delta\gamma_{\mathrm{ads}}$, Eq. (33)
C3 nucleate	R-slurry	heterogeneous barrier lowered, and accessible at Ω_{bulk}	$f(\theta)$ small; $\Delta G^{*}_{\mathrm{het}}(\Omega_{\mathrm{bulk}}) \le B_{\mathrm{acc}}$, Eqs. (24), (26)
C4 conformal	R-slurry, R-cell	a continuous film that closes below the thickness bound	$S_{\mathrm{spread}} \ge 0$ or closure below $d_{\max}$, Eqs. (37), (49)
C5 stable	R-slurry	$CaCO_3$ not dissolving; no solid below it; oxide baseline reported	$\Omega_{\mathrm{bulk}} \ge 1$ with the calcite K_{sp} (the Ω_{bulk} of C3), Eq. (43); ΔG_{pbx} within tolerance against solid competitors, Eq. (42)
C5 stable	R-cell	phases inside their Li-referenced window	$E_{\mathrm{red}} < E_{\mathrm{Li}} < E_{\mathrm{ox}}$, Eq. (45)

Table 1: Necessary conditions per link. C1–C3 are defined only in R-slurry: their reference states (water-wetted surfaces, hydrated Ca^{2+}, an aqueous ion activity product) do not exist in the cell.

Two rules follow from the regime split. First, C1–C3 make no statement about the cell: anchor retention, Ca^{2+} binding and nucleation in the organic electrolyte are not computed. Second, C5

needs two different constructions (Section 9), because aqueous Pourbaix analysis assumes water as the solvent and the hydrogen electrode as the reference.

2 Adhesion in water

Physics. In the slurry the template never meets a dry oxide. The hydroxylated Al_2O_3/AlOOH surface is covered by hydrogen-bonded water, and the polymer is hydrated too. A catechol–oxide contact can only form by displacing that water, so the relevant comparison is between two states that actually compete in the slurry: the oxide and polymer in contact, and the oxide and polymer each wetted by water. The dry (vacuum) work of adhesion compares the contact with two bare surfaces, a state that never occurs in R-slurry, and is therefore only a comparator.

The reversible work per unit area to separate a contact between phases A and B inside a medium M is the Dupré work $W = \gamma_{AM} + \gamma_{BM} - \gamma_{AB}$: the two new A–M and B–M interfaces are created and the A–B interface is destroyed. With A the oxide, B the polymer and M water,

$$W_{\mathrm{adh,wet}} = \gamma_{\mathrm{ow}} + \gamma_{\mathrm{pw}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]. \tag{1}$$

Units: each term is an interfacial free energy, so $W_{\mathrm{adh,wet}}$ is in $\mathrm{J\,m^{-2}}$. Writing every in-water energy through the dry works of each surface with water, $\gamma_{\mathrm{ow}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{w}} - W_{\mathrm{ow}}$ and $\gamma_{\mathrm{pw}} = \gamma_{\mathrm{p}} + \gamma_{\mathrm{w}} - W_{\mathrm{pw}}$, gives

$$W_{\mathrm{adh,wet}} = W_{\mathrm{adh,dry}} + 2\gamma_{\mathrm{w}} - W_{\mathrm{ow}} - W_{\mathrm{pw}}, \qquad W_{\mathrm{adh,dry}} = \gamma_{\mathrm{o}} + \gamma_{\mathrm{p}} - \gamma_{\mathrm{op}} \quad [\mathrm{J\,m^{-2}}]. \tag{2}$$

Equation (2) is the water-competition statement in one line: a large dry adhesion is offset by the works W_{ow} and W_{pw} with which water already binds each surface. A hydrophilic, hydroxylated oxide and a hydrophilic polymer bind water strongly, so $W_{\mathrm{adh,wet}}$ can be small or negative even when $W_{\mathrm{adh,dry}}$ is large. The bare-oxide control has $W_{\mathrm{adh,wet}} = 0$ by construction and fixes the zero of the scale.

The interfacial energies of Eq. (1) are not computed separately. The potential of mean force (PMF) of a template fragment along a separation coordinate in explicit water gives the free-energy difference between the separated and bound states, ΔF_{PMF} (J per fragment), which already contains the displacement penalty and the entropy of the released water. Normalising by the contact area,

$$W_{\mathrm{adh,wet}} = \frac{\Delta F_{\mathrm{PMF}}}{A_{\mathrm{c}}} \quad [\mathrm{J\,m^{-2}}], \tag{3}$$

with units $\mathrm{J/m^2}$. Two conditions make Eq. (3) meaningful: the PMF must reach a plateau at large separation (both surfaces fully rewetted), and A_{c} must be defined once for all fragments, because the ranking across binders depends on it (open question Q5). C1 requires $W_{\mathrm{adh,wet}} > 0$ beyond its uncertainty on the populated oxide terminations.

3 The sequestration cycle and the enrichment factor

Physics. The template can raise the local Ca^{2+} concentration only if it holds Ca^{2+} more strongly than water does. The question is a competition between two ligand environments for the same ion: the hydration shell in bulk solution and the coordinating groups of the template. The free energy must therefore start from hydrated Ca^{2+} at the regime activity, include the cost of preparing the binding site at the regime pH, and end in a well-defined local state.

Cycle and standard states. Let the coordinating group be an acid L–H that binds Ca^{2+} only when deprotonated. The cycle is

$$\text{(i)}\quad \mathrm{L{-}H\ (layer)} \rightarrow \mathrm{L^-(layer) + H^+(aq)}, \qquad \Delta G_{\mathrm{i}} = -RT\ln\alpha \quad [\mathrm{J\,mol^{-1}}]$$

$$\text{(ii)}\quad \mathrm{Ca^{2+}(aq,\ bulk) + L^-} \rightarrow \mathrm{Ca^{2+}\cdot L^-(layer)}, \qquad \Delta G_{\mathrm{ii}} = \Delta G_{\mathrm{site}} \quad [\mathrm{J\,mol^{-1}}]$$

with $\alpha = 1/\big(1 + 10^{\,\mathrm{p}K_\mathrm{a} - \mathrm{pH}}\big)$ (dimensionless). Step (i) is the free energy of finding the site deprotonated when the solution fixes the proton activity: α is the deprotonated fraction at the regime pH, and $-RT \ln \alpha \geq 0$. Step (ii) is the transfer of one Ca^{2+} from bulk solution into the binding layer. Summing,

$$\Delta G_\mathrm{seq} = \Delta G_\mathrm{site} + RT \ln\big(1 + 10^{\,\mathrm{p}K_\mathrm{a} - \mathrm{pH}}\big) \quad [\mathrm{J\,mol^{-1}}]. \tag{4}$$

Units: RT is in $\mathrm{J\,mol^{-1}}$ and the logarithm is of a pure number. The correction vanishes as $\mathrm{pH} - \mathrm{p}K_\mathrm{a} \to \infty$, equals $RT \ln 2$ at $\mathrm{pH} = \mathrm{p}K_\mathrm{a}$, and grows as the site becomes protonated. Both $\mathrm{p}K_\mathrm{a}$ and the activity coefficients are taken at the regime ionic strength. This is why catechol, largely protonated near neutral pH, must be tested rather than assumed (design flag 4).

Step (ii) fixes the standard states. Let $w(\mathbf{r})$ be the PMF of one Ca^{2+} relative to bulk solution (J per ion, $w \to 0$ far from the template), computed with counter-ions and water explicit. In the dilute limit the equilibrium concentration is $c(\mathbf{r}) = c_\mathrm{bulk}\, \mathrm{e}^{-w(\mathbf{r})/k_\mathrm{B}T}$. Assigning to each site a volume V_site (the binding-layer volume per site) and defining

$$\Delta G_\mathrm{site} \equiv -RT \ln\Big[\frac{1}{V_\mathrm{site}} \int_\mathrm{site} \mathrm{e}^{-w(\mathbf{r})/k_\mathrm{B}T}\, \mathrm{d}^3 r\Big] \quad [\mathrm{J\,mol^{-1}}] \tag{5}$$

(the bracket is a ratio of volumes, so it is a pure number) makes the transfer free energy a partition between two concentration scales: the bulk at the regime activity on the solution side, and the mean concentration in the site volume on the layer side. The conventional binding free energy to a 1 mol-per-volume standard state c° differs from it by the standard-state correction

$$\Delta G^\circ_\mathrm{bind} = \Delta G_\mathrm{site} - RT \ln\big(N_\mathrm{A} c^\circ V_\mathrm{site}\big) \quad [\mathrm{J\,mol^{-1}}], \tag{6}$$

where $N_\mathrm{A} c^\circ V_\mathrm{site}$ is dimensionless. A literature binding constant can therefore be compared with ΔG_site only after Eq. (6), and only with the same V_site. The choice of V_site shifts ΔG_site by $RT \ln(V'_\mathrm{site}/V_\mathrm{site})$, so it is fixed once (open question Q4).

Activity versus concentration. Equation (5) defines a concentration ratio. Because w contains every interaction of the ion with its surroundings, the electrochemical potential of Ca^{2+} is uniform at equilibrium (Section 4): the ion in the layer carries the same chemical potential as in bulk, and the whole enrichment sits in its local standard potential, equivalently in a local activity coefficient $\gamma_\mathrm{loc} = \gamma_\mathrm{bulk}/\mathcal{E}$. $\mathcal{E}$ is therefore a concentration ratio and never an activity ratio. Contract v1 used a *local-ideal* bookkeeping that gave the enriched ions the bulk standard state; Section 7 shows that it is inconsistent at equilibrium and replaces it.

Enrichment and its validity range. In the dilute limit with non-interacting sites,

$$\mathcal{E} \equiv \frac{c_\mathrm{local}}{c_\mathrm{bulk}} = \exp\Big(-\frac{\Delta G_\mathrm{seq}}{RT}\Big) \quad [1]. \tag{7}$$

$\Delta G_\mathrm{seq} < 0$ gives enrichment ($\mathcal{E} > 1$); $\Delta G_\mathrm{seq} > 0$ gives depletion: the template is then a worse ligand than water. Equation (7) holds when (a) Ca^{2+} is dilute in bulk, (b) the sites are far from saturation, (c) one ion binds per site and sites do not interact, and (d) the non-specific (ionic-strength) activity coefficient is the same in the layer as in bulk, so that $\mathcal{E}$ is fixed by the PMF alone. Condition (b) is quantified next.

Saturation bound. The layer has Γ_s sites per unit area in a thickness δ_s, so its Ca^{2+} content cannot exceed $c_\mathrm{max} = \Gamma_\mathrm{s}/\delta_\mathrm{s}$ ($\mathrm{mol\,m^{-2}}$ over m gives $\mathrm{mol\,m^{-3}}$). For independent, equivalent sites (Langmuir), the occupied fraction is $\Theta = x/(1+x)$ with $x = \mathcal{E} c_\mathrm{bulk}/c_\mathrm{max}$, and $c_\mathrm{local} = c_\mathrm{max}\Theta$:

$$c_\mathrm{local} = \frac{\mathcal{E} c_\mathrm{bulk}}{1 + \mathcal{E} c_\mathrm{bulk}/c_\mathrm{max}} \leq \min\big(\mathcal{E} c_\mathrm{bulk},\ c_\mathrm{max}\big) \quad [\mathrm{mol\,m^{-3}}]. \tag{8}$$

The dilute result (7) is the limit $x \ll 1$; at $x \gg 1$ the enrichment saturates at $c_{\max}/c_{\text{bulk}}$ however negative ΔG_{seq} is. A strongly binding but sparse template is therefore bounded by its site density, not by its binding free energy. Electrostatic coupling between charged sites, which Langmuir ignores, is open question Q3.

Occupancy and surface excess. The same closure gives the fraction of occupied sites and the Gibbs relative excess of Ca^{2+} at the template–liquid interface, the bound amount per area less the amount the layer volume would hold at the bulk concentration:

$$\Theta = \frac{x}{1+x} = \frac{c_{\text{local}}}{c_{\max}}, \qquad x = \frac{\mathcal{E} c_{\text{bulk}}}{c_{\max}} \quad [1], \qquad \Gamma_{\text{Ca}} = \Gamma_{\text{s}}\Theta - \delta_{\text{s}} c_{\text{bulk}} \quad [\text{mol m}^{-2}]. \tag{9}$$

x is the occupancy odds of a site at the regime activity. $\Gamma_{\text{Ca}} > 0$ for net binding and $\Gamma_{\text{Ca}} < 0$ for a template that repels Ca^{2+}. These, not a local supersaturation, are what C2 hands to C3 (Section 7).

4 The driving force at the template

Physics. A nucleus is built from ions taken out of the solution, and the free energy it gains per formula unit is the chemical potential those ions carry minus that of the solid. The template sits in an open interfacial region that exchanges ions with the slurry. At equilibrium nothing can be gained by moving an ion between the bulk and the layer, so every ion carries the same electrochemical potential everywhere: the template can concentrate ions, but it cannot make them carry more free energy. The driving force at the template is therefore the bulk one. The algebra below shows this for the two ways a template enriches Ca^{2+}, specific binding and a mean electrostatic potential.

One definition. Throughout,

$$\Omega = \frac{\text{IAP}}{K_{\text{sp}}} = \frac{a_{\text{Ca}^{2+}}\, a_{\text{CO}_3^{2-}}}{K_{\text{sp}}} \quad [1], \qquad \Delta\mu = k_{\text{B}} T \ln \Omega \quad [\text{J}], \tag{10}$$

with K_{sp} the solubility product of the framing (calcite and ACC are both registered, and each is run as a whole chain with its own K_{sp}, V_{m} and γ_{nl}), $\Delta\mu$ the chemical-potential drop per $CaCO_3$ formula unit on crystallisation, and V_{m} the molar volume per $CaCO_3$ formula unit. Units: activities and K_{sp} are pure numbers on the molal scale; $k_{\text{B}}T$ is in J. Nucleation requires $\Omega > 1$. This is the charter's convention (F3): its S is this Ω. The per-ion square root $\Omega^{1/2}$ with a factor $2k_{\text{B}}T$ is the same statement and is not used, so that no factor of two can be lost between lanes.

Structural water (Q13). For a hydrated framing $CaCO_3 \cdot nH_2O$ the ion activity product is $a_{\text{Ca}^{2+}} a_{\text{CO}_3^{2-}} a_{\text{w}}^{n}$, with K_{sp} of the same framing. The screen keeps IAP $= a_{\text{Ca}^{2+}} a_{\text{CO}_3^{2-}}$. In the slurry $a_{\text{w}} = 1$ and the two agree. At a drying state $a_{\text{w}} < 1$, and the anhydrous form overstates Ω by the factor $a_{\text{w}}^{-n} > 1$: the drying states are optimistic, biased toward a larger driving force. The screen therefore reports a_{w} at every drying state and, where n is supplied, the hydrated ratio as a labelled sensitivity (`Omega_bulk_hydrated`); it never decides a verdict (cp2 fix F7).

Activities from speciation. Ω needs the activities of the *free* ions, and in an alkaline, carbonate-bearing slurry part of the calcium and carbon is bound in ion pairs. The registered state fixes pH, the dissolved inorganic carbon C_{T}, the total dissolved calcium Ca_{T} and the ionic strength I; the solution is closed (no CO_2 exchange) and not yet precipitated. With $a_{\text{H}} = 10^{-\text{pH}}$, the species are tied to $y \equiv a_{\text{CO}_3^{2-}}$ and $a_{\text{Ca}^{2+}}$ by the thermodynamic constants (all at $I = 0$): $a_{\text{HCO}_3} = 10^{\text{p}K_2 - \text{pH}} y$, $a_{\text{CO}_2} = 10^{\text{p}K_1 + \text{p}K_2 - 2\text{pH}} y$, $a_{\text{CaCO}_3^0} = K_{\text{CaCO}_3^0} a_{\text{Ca}^{2+}} y$, $a_{\text{CaHCO}_3^+} = K_{\text{CaHCO}_3^+} a_{\text{Ca}^{2+}} a_{\text{HCO}_3}$ and $a_{\text{CaOH}^+} = {}^{*}K_{\text{CaOH}^+} a_{\text{Ca}^{2+}} a_{\text{w}} / a_{\text{H}}$. Dividing each activity by its activity coefficient ($\gamma_{\pm 1}$, $\gamma_{\pm 2}$ by

charge, one for neutral species) and summing gives the two mass balances, in units of the standard molality m°:

$$m_{C_\mathrm{T}} = A_\mathrm{c}\, y + P_\mathrm{c}\, a_{\mathrm{Ca}^{2+}} y, \qquad m_{\mathrm{Ca}_\mathrm{T}} = D_\mathrm{c}\, a_{\mathrm{Ca}^{2+}} + P_\mathrm{c}\, a_{\mathrm{Ca}^{2+}} y \quad [1], \tag{11}$$

with $A_\mathrm{c} = 10^{\mathrm{p}K_1+\mathrm{p}K_2-2\mathrm{pH}} + 10^{\mathrm{p}K_2-\mathrm{pH}}/\gamma_{\pm1} + 1/\gamma_{\pm2}$ (free carbon per unit carbonate activity), $P_\mathrm{c} = K_{\mathrm{CaCO}_3^0} + K_{\mathrm{CaHCO}_3^+} 10^{\mathrm{p}K_2-\mathrm{pH}}/\gamma_{\pm1}$ (ion pairs per unit $a_{\mathrm{Ca}^{2+}} y$), $D_\mathrm{c} = 1/\gamma_{\pm2} + {}^*K_{\mathrm{CaOH}^+} a_\mathrm{w} 10^{\mathrm{pH}}/\gamma_{\pm1}$ (free calcium per unit $a_{\mathrm{Ca}^{2+}}$), and $m_{C_\mathrm{T}} = C_\mathrm{T}/(\rho_\mathrm{w} m^\circ)$, $m_{\mathrm{Ca}_\mathrm{T}} = \mathrm{Ca}_\mathrm{T}/(\rho_\mathrm{w} m^\circ)$ (dilute solution). Eliminating $a_{\mathrm{Ca}^{2+}} = m_{\mathrm{Ca}_\mathrm{T}}/(D_\mathrm{c} + P_\mathrm{c} y)$ leaves $A_\mathrm{c} P_\mathrm{c} y^2 + \big[A_\mathrm{c} D_\mathrm{c} + P_\mathrm{c}(m_{\mathrm{Ca}_\mathrm{T}} - m_{C_\mathrm{T}})\big] y - m_{C_\mathrm{T}} D_\mathrm{c} = 0$. The product of its roots is $-m_{C_\mathrm{T}} D_\mathrm{c}/(A_\mathrm{c} P_\mathrm{c}) < 0$, so exactly one root is positive:

$$a_{\mathrm{CO}_3^{2-}} = \frac{2 m_{C_\mathrm{T}} D_\mathrm{c}}{B + \sqrt{B^2 + 4 A_\mathrm{c} P_\mathrm{c} m_{C_\mathrm{T}} D_\mathrm{c}}}, \qquad B = A_\mathrm{c} D_\mathrm{c} + P_\mathrm{c}(m_{\mathrm{Ca}_\mathrm{T}} - m_{C_\mathrm{T}}) \quad [1], \tag{12}$$

written in the form that stays exact as $P_\mathrm{c} \to 0$. The limit cases are checked in the contract (`a_CO3_speciation`): with no ion pairs, or with no calcium ($m_{\mathrm{Ca}_\mathrm{T}} = 0$, where the root becomes $\sqrt{(A_\mathrm{c} D_\mathrm{c} + P_\mathrm{c} m_{C_\mathrm{T}})^2}$), $a_{\mathrm{CO}_3^{2-}} = m_{C_\mathrm{T}}/A_\mathrm{c}$, the carbonate system alone; pairs can only lower it. The free Ca^{2+} concentration $c_\mathrm{bulk} = a_{\mathrm{Ca}^{2+}} \rho_\mathrm{w} m^\circ/\gamma_{\pm2}$ is the c_bulk of the Langmuir and Gibbs relations of Section 3: the free ion, not the total.

Activity model. The Davies equation, $\log_{10}\gamma_{\pm z} = -A_\mathrm{DH} z^2 \big[s/(1+s) - b_\mathrm{D} s^2\big]$ with $s = \sqrt{I/m^\circ}$, so $\gamma_{\pm2} = \gamma_{\pm1}^4$. As $I \to 0$ every coefficient tends to one and the activities are ideal (checked). A_DH (at the regime temperature) and b_D are fetched, as is the registered validity bound $I \le I_\mathrm{val}$. Independently of that bound, $\mathrm{d}\log_{10}\gamma/\mathrm{d}s = -A_\mathrm{DH} z^2 \big[(1+s)^{-2} - 2 b_\mathrm{D} s\big]$ changes sign at the one positive root s_* of $2 b_\mathrm{D} s (1+s)^2 = 1$; beyond $I_* = m^\circ s_*^2$ the ionic-strength dependence is carried by the empirical linear term alone, so $I_\mathrm{val} \le I_*$ is required. A draw with $I > I_\mathrm{val}$ is refused and counted, never extrapolated. Neutral species are taken with activity coefficient one (no salting term), and I is the registered input rather than recomputed from the speciation; both are stated limitations.

Equilibrium of the open interfacial region. Write the electrochemical potential of ion i (charge number z_i) at position $\mathbf{r}$ as

$$\tilde{\mu}_i(\mathbf{r}) = \mu_i^\circ + w_i^\mathrm{sp}(\mathbf{r}) + k_\mathrm{B} T \ln \frac{\gamma_i(\mathbf{r})\, m_i(\mathbf{r})}{m^\circ} + z_i e \psi(\mathbf{r}) \quad [\mathrm{J}], \tag{13}$$

where μ_i° is the bulk standard potential, w_i^sp the specific (non-electrostatic) interaction with the template ($w^\mathrm{sp} \to 0$ in bulk), γ_i the non-specific activity coefficient, m_i the molality and ψ the mean electrostatic potential ($\psi \to 0$ in bulk). Equilibrium of the region with the slurry, which acts as a reservoir, is the statement

$$\tilde{\mu}_i(\mathbf{r}) = \tilde{\mu}_{i,\mathrm{bulk}} \quad \text{for every ion } i \text{ and every } \mathbf{r}. \tag{14}$$

Solving Eq. (13) for m_i gives the local enrichment, $m_i(\mathbf{r})/m_{i,\mathrm{bulk}} = (\gamma_{i,\mathrm{bulk}}/\gamma_i) \exp[-(w_i^\mathrm{sp} + z_i e\psi)/k_\mathrm{B}T]$: the Boltzmann factor of Section 3 is a *consequence* of Eq. (14), not an exception to it.

The driving force for adding one formula unit to a nucleus at $\mathbf{r}$ is

$$\Delta\mu(\mathbf{r}) = \tilde{\mu}_\mathrm{Ca}(\mathbf{r}) + \tilde{\mu}_{\mathrm{CO}_3}(\mathbf{r}) - \mu_{\mathrm{CaCO}_3(\mathrm{s})} = \tilde{\mu}_{\mathrm{Ca,bulk}} + \tilde{\mu}_{\mathrm{CO}_3,\mathrm{bulk}} - \mu_{\mathrm{CaCO}_3(\mathrm{s})} = k_\mathrm{B} T \ln \Omega_\mathrm{bulk} \quad [\mathrm{J}]. \tag{15}$$

It does not depend on $\mathbf{r}$: it is the same at a binding site, in the diffuse layer and in the bulk.

(i) Specific binding. At a site, $w_\mathrm{Ca}^\mathrm{sp}$ adds $\Delta G_\mathrm{seq}/N_\mathrm{A}$ per ion to the standard part of $\tilde{\mu}_\mathrm{Ca}$ (a lowering when $\Delta G_\mathrm{seq} < 0$) (with the site-volume standard state of Eq. (5)) and raises the concentration term by $k_\mathrm{B} T \ln \mathcal{E} = -\Delta G_\mathrm{seq}/N_\mathrm{A}$. The two changes are equal and opposite, which is why $\tilde{\mu}_\mathrm{Ca}$ stays at its bulk value. The local enrichment $c_\mathrm{local}/c_\mathrm{bulk} = \mathcal{E} = \exp(-\Delta G_\mathrm{seq}/RT)$ therefore changes the driving force by

$$\Delta\mu_\mathrm{site} - \Delta\mu_\mathrm{bulk} = \underbrace{k_\mathrm{B} T \ln \mathcal{E}}_{\text{concentration}} + \underbrace{\Delta G_\mathrm{seq}/N_\mathrm{A}}_{\text{standard potential}} = 0 \quad [\mathrm{J}]. \tag{16}$$

(ii) A mean potential. At an anionic template ($\psi < 0$) Ca^{2+} is enriched by $e^{-2e\psi/k_BT}$ and CO_3^{2-} depleted by $e^{+2e\psi/k_BT}$. Even counting concentrations alone, the ion product is unchanged, $(a_{Ca^{2+}}a_{CO_3^{2-}})_{local} = (a_{Ca^{2+}}a_{CO_3^{2-}})_{bulk}$, and in Eq. (15) the potential enters as $(z_{Ca} + z_{CO_3})e\psi = 0$ because the solid is neutral. A mean potential changes neither Ω nor $\Delta\mu$. This settles Q2: carbonate needs no separate treatment, and holding it at its bulk activity (the v1 choice) is not the issue; crediting the enriched Ca^{2+} was.

Result. At equilibrium the nucleus at the template is driven by $k_BT \ln \Omega_{bulk}$, evaluated at the regime activities. The template changes the barrier only through the interfacial energies: the diffuse-layer free energy is part of γ_{sl} and γ_{nl}, and the specific binding enters γ_{sl} through the Gibbs adsorption equation (Section 7). The local-ideal ratio $\Omega_{li} = (c_{local}/c_{bulk})\Omega_{bulk}$, with $\Delta\mu_{li} = \Delta\mu_{bulk} - \Delta G_{seq}/N_A$, survives only as the named double-counting demonstration of the contract (`Omega_local_ideal`, `Delta_mu_local_ideal`).

5 Homogeneous and heterogeneous nucleation

Physics. A cluster of the new phase gains $\Delta\mu$ for every formula unit it contains but pays γ for every unit of interface it creates. Volume grows as r^3 and surface as r^2, so small clusters are unstable and large ones grow; the barrier is the free energy at the crossover. On a substrate the cluster can replace part of its interface with a cheaper substrate contact, which lowers the barrier without changing the driving force.

Homogeneous barrier. In the capillarity approximation, a spherical cluster of radius r with molecular volume $v_m = V_m/N_A$ has

$$\Delta G_{hom}(r) = -\frac{4\pi r^3}{3}\frac{\Delta\mu}{v_m} + 4\pi r^2\gamma_{nl} \quad [\mathrm{J}]. \tag{17}$$

Units: r^3/v_m counts formula units, times J per formula unit; $r^2\gamma_{nl}$ is m^2 times $J\,m^{-2}$. Setting $\partial\Delta G_{hom}/\partial r = 0$,

$$r^* = \frac{2\gamma_{nl}v_m}{\Delta\mu} \quad [\mathrm{m}], \qquad \Delta G^*_{hom} = \frac{16\pi\gamma_{nl}^3 v_m^2}{3\Delta\mu^2} = \frac{4\pi r^{*2}\gamma_{nl}}{3} \quad [\mathrm{J}]. \tag{18}$$

Units: $J\,m^{-2}\cdot m^3/J = m$ for r^*; $J^3\,m^{-6}\cdot m^6/J^2 = J$ for ΔG^*_{hom}. With $\Delta\mu = k_BT\ln\Omega$, $\Delta G^*_{hom} \to \infty$ as $\Omega \to 1^+$ (no driving force, no nucleation) and $\Delta G^*_{hom} \to 0$ as $\Omega \to \infty$; it increases with γ_{nl} and V_m and decreases with T at fixed Ω. The absolute barrier in units of k_BT is $\Delta G^*_{hom}/k_BT$.

Spherical-cap geometry. On a flat substrate the cluster is a cap cut from a sphere of radius r, meeting the substrate at contact angle θ measured through the cap. Elementary geometry gives base radius $\rho = r\sin\theta$, height $h = r(1-\cos\theta)$, and

$$A_{cap} = 2\pi r^2(1-\cos\theta), \qquad A_{base} = \pi r^2\sin^2\theta \quad [\mathrm{m}^2], \tag{19}$$

$$V_{cap} = \frac{\pi r^3}{3}(1-\cos\theta)^2(2+\cos\theta) \quad [\mathrm{m}^3]. \tag{20}$$

The cap's free energy has three parts: the volume term, the free nucleus–liquid surface, and the base, where substrate–liquid interface (γ_{sl}) is replaced by substrate–nucleus interface (γ_{sn}):

$$\Delta G_{het}(r) = -V_{cap}\frac{\Delta\mu}{v_m} + \gamma_{nl}A_{cap} + (\gamma_{sn}-\gamma_{sl})A_{base} \quad [\mathrm{J}]. \tag{21}$$

Stationarity of Eq. (21) with respect to θ at fixed cap volume is the contact-line balance, Young's relation $\gamma_{sl} - \gamma_{sn} = \gamma_{nl}\cos\theta$ (Section 6). Substituting it, the surface terms become

$$\gamma_{nl}\pi r^2\left[2(1-\cos\theta) - \cos\theta\sin^2\theta\right] = \gamma_{nl}\pi r^2(1-\cos\theta)^2(2+\cos\theta) = 4\pi r^2\gamma_{nl}\,f(\theta) \quad [\mathrm{J}], \tag{22}$$

using $2 - \cos\theta - \cos^2\theta = (1 - \cos\theta)(2 + \cos\theta)$, and the volume term is $\frac{4\pi r^3}{3} f(\theta)$ times $-\Delta\mu/v_{\mathrm{m}}$, with

$$f(\theta) = \frac{(2 + \cos\theta)(1 - \cos\theta)^2}{4} = \frac{2 - 3\cos\theta + \cos^3\theta}{4} \quad [1]. \tag{23}$$

Hence $\Delta G_{\mathrm{het}}(r) = f(\theta)\,\Delta G_{\mathrm{hom}}(r)$ at every r: the cap has the same critical radius as the sphere, and

$$\Delta G^*_{\mathrm{het}} = f(\theta)\,\Delta G^*_{\mathrm{hom}} \quad [\mathrm{J}]. \tag{24}$$

The factorisation holds only because one and the same γ_{nl} multiplies A_{cap} and enters Young's relation; this is the reason Section 6 keeps the bare γ_{nl} in Young.

Properties of f. $f(0) = 0$ (complete wetting, no barrier), $f(\pi/2) = \frac{1}{2}$ (hemisphere), $f(\pi) = 1$ (no wetting, the homogeneous barrier). Its derivative is $f'(\theta) = \frac{3}{4}\sin^3\theta \geq 0$ on $[0, \pi]$, so f is monotone and $0 \leq \Delta G^*_{\mathrm{het}} \leq \Delta G^*_{\mathrm{hom}}$ for every admissible θ. f is also the volume fraction of the cap relative to the full sphere of the same radius.

Assumptions. Capillarity (macroscopic interfacial energies at nucleus size), an isotropic γ_{nl} (closer to the amorphous framing than to faceted calcite), a flat, homogeneous substrate, no line tension, and a cluster of the bulk phase's composition. A non-classical pathway (prenucleation clusters, liquid precursors) is outside this framework and is a limitation, not a parameter.

Accessibility of the barrier. A barrier is low enough only relative to the number of attempts the system makes in the time available. Nuclei form on the template at the rate per unit area

$$J = J_0\,\mathrm{e}^{-\Delta G^*_{\mathrm{het}}/k_{\mathrm{B}}T} \quad [\mathrm{m}^{-2}\,\mathrm{s}^{-1}], \tag{25}$$

with J_0 the kinetic prefactor of heterogeneous nucleation per unit template area (attachment frequency, Zeldovich factor and site density). Over the time t_{p} that the template spends at the registered R-slurry state (slurry contact and drying), the expected areal density of nuclei is $N_{\mathrm{nuc}} = J\,t_{\mathrm{p}}$; along a path of registered states k held for Δt_k it is $\sum_k J_0 \Delta t_k\,\mathrm{e}^{-\Delta G^*_{\mathrm{het},k}/k_{\mathrm{B}}T}$. Requiring at least the areal density N_{req} gives the absolute-barrier criterion

$$N_{\mathrm{nuc}} \geq N_{\mathrm{req}} \iff \Delta G^*_{\mathrm{het}} \leq B_{\mathrm{acc}} = k_{\mathrm{B}}T \ln\frac{J_0\,t_{\mathrm{p}}}{N_{\mathrm{req}}} \quad [\mathrm{J}]. \tag{26}$$

Units: $J_0 t_{\mathrm{p}}$ and N_{req} are both m^{-2}, so the logarithm is of a pure number. Equivalently, a count $\mathcal{N}$ on a collector area A is $N_{\mathrm{req}} = \mathcal{N}/A$. What must be required is fixed by the contact patch A_{patch} through which a cathode particle touches the collector: a patch with no nucleus under it is a patch the mineral cannot cover from below. If nuclei land independently at mean areal density N_{nuc}, the number under one patch is Poisson distributed with mean $N_{\mathrm{nuc}}A_{\mathrm{patch}}$, and the fraction of patches left empty is $\mathrm{e}^{-N_{\mathrm{nuc}}A_{\mathrm{patch}}}$. Requiring that at most a fraction q_0 of the patches be empty gives

$$\mathrm{e}^{-N_{\mathrm{nuc}}A_{\mathrm{patch}}} \leq q_0 \iff N_{\mathrm{nuc}} \geq N_{\mathrm{req}} = \frac{\ln(1/q_0)}{A_{\mathrm{patch}}} \quad [\mathrm{m}^{-2}], \tag{27}$$

and P0 registers $q_0 = 1 - p$, with p the credible level of the decision rule (`theory/thresholds.md`; cp3). The expectation form, one nucleus per patch on average, is the special case $q_0 = 1/\mathrm{e}$. N_{req} is fixed separately from the closure density of C4 (Section 8), so that C3 does not borrow the island geometry of C4. Four properties make the criterion usable without a precise kinetic theory: only $\ln J_0$ enters, so an order-of-magnitude prefactor moves B_{acc} by a few $k_{\mathrm{B}}T$; the barrier is evaluated at Ω_{bulk} (Section 7); the non-stationary time lag is neglected; and N_{nuc} ignores exclusion zones around growing islands, so it is an upper estimate and the criterion is necessary, not sufficient. It is a statement about nucleation in the slurry, not about corrosion, and no rate of anything else follows from it.

6 Young's relation and the fixed sign convention

Physics. At the contact line, moving the line outward by a small amount replaces substrate–liquid interface by substrate–nucleus interface and stretches the nucleus–liquid surface. At equilibrium these balance. The contact angle therefore measures how much the substrate prefers the nucleus to the liquid, in units of the nucleus–liquid energy.

Convention, in words. θ is the angle at the three-phase line measured through the nucleus. $\cos\theta$ is the free energy gained per unit contact area by replacing substrate–liquid interface with substrate–nucleus interface, divided by the nucleus–liquid interfacial free energy. $\theta = 0$ means the nucleus wets the template completely; $\theta = \pi$ means it does not wet it at all.

Convention, as an equation.

$$\cos\theta = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}}{\gamma_{\mathrm{nl}}}, \qquad \gamma_{\mathrm{sn}} = \gamma_{\mathrm{sn}}^{(0)} + \Delta\gamma_{\mathrm{w}} \quad [1;\ \mathrm{J\,m^{-2}}]. \tag{28}$$

Units: a ratio of interfacial free energies. A finite angle exists only for $|\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}| \le \gamma_{\mathrm{nl}}$. For $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} \ge \gamma_{\mathrm{nl}}$ the nucleus phase spreads ($S_{\mathrm{spread}} \ge 0$, Section 8) and $f = 0$; for $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} \le -\gamma_{\mathrm{nl}}$ it does not wet and $f = 1$. An implementation clips $\cos\theta$ to $[-1, 1]$ and reports which branch applied.

The water-displacement term. The template is hydrated. When the nucleus forms on it, the template-bound water layer under the contact disc must be displaced, and that costs free energy over the contact area only. Define

$$\Delta\gamma_{\mathrm{w}} \equiv \gamma_{\mathrm{sn}} - \gamma_{\mathrm{sn}}^{(0)} \quad [\mathrm{J\,m^{-2}}], \tag{29}$$

where $\gamma_{\mathrm{sn}}^{(0)}$ is a substrate–nucleus energy from a reference that does not include the removal of that layer (for example, a contact computed without the template's structured water). The sign convention is: $\Delta\gamma_{\mathrm{w}} > 0$ **is a cost.** From Eq. (28), $\partial\cos\theta/\partial\Delta\gamma_{\mathrm{w}} = -1/\gamma_{\mathrm{nl}} < 0$, and with $\partial f/\partial\cos\theta = -\frac{3}{4}\sin^2\theta \le 0$,

$$\frac{\partial f}{\partial\Delta\gamma_{\mathrm{w}}} = \frac{3\sin^2\theta}{4\gamma_{\mathrm{nl}}} \ge 0 \quad [\mathrm{m^2\,J^{-1}}]: \tag{30}$$

a positive displacement cost raises θ, f and $\Delta G^*_{\mathrm{het}}$, and $\Delta\gamma_{\mathrm{w}} < 0$ lowers them. If γ_{sn} is supplied directly as a full in-water interfacial free energy, the displacement is already inside it, $\Delta\gamma_{\mathrm{w}} \equiv 0$, and the term must not be added again. Applying it twice is the double-counting error that C3 item 13 demonstrates.

Why Young uses the bare γ_{nl} and not γ_{eff}. The deck writes $\gamma_{\mathrm{eff}} = \gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}}$ and places it in the denominator of Young. That is rejected, for three reasons. (1) Geometry: the displaced water sits on the contact disc. The free surface of the cap, whose energy is $\gamma_{\mathrm{nl}}A_{\mathrm{cap}}$, never touches the template, so nothing about the template's hydration changes γ_{nl}. (2) Consistency: the factorisation of Eq. (24) requires the γ_{nl} in Young to equal the γ_{nl} that multiplies A_{cap} and sets $\Delta G^*_{\mathrm{hom}}$. A modified denominator would leave a residual surface term and break $\Delta G^*_{\mathrm{het}} = f\Delta G^*_{\mathrm{hom}}$. (3) Sign: with γ_{eff} in the denominator,

$$\left.\frac{\partial\cos\theta}{\partial\Delta\gamma_{\mathrm{w}}}\right|_{\mathrm{deck}} = \frac{\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}^{(0)}}{(\gamma_{\mathrm{nl}} - \Delta\gamma_{\mathrm{w}})^2} \quad [\mathrm{m^2\,J^{-1}}], \tag{31}$$

which is positive on the wetting side ($\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}^{(0)}$): a displacement *cost* would lower the barrier. This is design flag 3. γ_{eff} is kept only as a named diagnostic, so that W4 can reproduce the deck's form for the double-counting demonstration. The contract encodes the reversal as sign-flip test SF1 (Table 2).

Test, change	Registered convention (contract v2)	Rejected form
SF1, $\Delta\gamma_{\mathrm{w}} > 0$	$\cos\theta \downarrow$, $\Delta G^{*}_{\mathrm{het}} \geq \Delta G^{*}_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$	deck (γ_{eff} in the denominator), $\gamma_{\mathrm{sl}} > \gamma^{(0)}_{\mathrm{sn}}$: $\cos\theta \uparrow$, $\Delta G^{*}_{\mathrm{het}} \leq \Delta G^{*}_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$
SF1, $\Delta\gamma_{\mathrm{w}} < 0$	$\Delta G^{*}_{\mathrm{het}} \leq \Delta G^{*}_{\mathrm{het}}\vert_{\Delta\gamma_{\mathrm{w}}=0}$	reversed
SF2, $\Delta G_{\mathrm{seq}} \downarrow$	$\Omega = \Omega_{\mathrm{bulk}}$; $\cos\theta \downarrow$ through $\Delta\gamma_{\mathrm{ads}}$; barrier raised at fixed $\gamma^{(0)}_{\mathrm{sl}}, \gamma_{\mathrm{sn}}$	local-ideal (v1): $\Omega_{\mathrm{local}} > \Omega_{\mathrm{bulk}}$, barrier lowered
SF3, $\gamma_{\mathrm{nl}} \downarrow$	$f \downarrow$ if $\gamma_{\mathrm{sl}} > \gamma_{\mathrm{sn}}$; $f \uparrow$ if $\gamma_{\mathrm{sl}} < \gamma_{\mathrm{sn}}$; $f = \frac{1}{2}$ fixed if equal	"lowering γ_{nl} always lowers f"

Table 2: Sign-flip tests SF1–SF3 of `theory/contract.yaml`.

7 Where the template acts: bookkeeping and the no-double-counting proof

Physics. The template is both an interface and a ligand field. In classical nucleation the barrier depends on four quantities, $(\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sn}}, \gamma_{\mathrm{nl}}, \Delta\mu)$; γ_{nl} belongs to the nucleus and the solution, so the template can act only through γ_{sl}, γ_{sn} and $\Delta\mu$. Section 4 showed that at equilibrium it cannot act through $\Delta\mu$. What remains is to find where its Ca^{2+} binding goes, to show that it is counted exactly once, and to state the rule the screen follows.

7.1 Two bookkeepings

Uniform chemical potential (U). Every ion carries its bulk electrochemical potential, Eq. (14); the driving force is $k_{\mathrm{B}}T \ln \Omega_{\mathrm{bulk}}$, Eq. (15); and every interaction between the template and the ions near it is an excess quantity per area inside the grand-canonical interfacial energies γ_{sl} and γ_{sn}, evaluated at the bulk chemical potentials.

Local-ideal (L). The enriched Ca^{2+} is given the bulk standard state at the local concentration, so the enrichment raises the supersaturation, $\Omega_{\mathrm{li}} = (c_{\mathrm{local}}/c_{\mathrm{bulk}})\Omega_{\mathrm{bulk}}$, and the interfacial energies are taken for the template without its Ca^{2+} excess. This was contract v1.

Which is consistent. (U) is the equilibrium statement. (L) makes a hidden assumption: an ion in the layer carries $\tilde{\mu}_{\mathrm{Ca,bulk}} - \Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$, more than the bulk ion when $\Delta G_{\mathrm{seq}} < 0$. That is the electrochemical potential of a layer that is *not* in equilibrium with the slurry, a separate reservoir held above the bulk. (L) is therefore self-contradictory: $\mathcal{E}$ is computed from the equilibrium condition (14) and then used as if that condition did not hold. The binding free energy is spent once to hold the ion at the site and credited a second time as driving force.

When a local excess of driving force is real. A genuine $\Delta\mu_{\mathrm{local}} > \Delta\mu_{\mathrm{bulk}}$ needs a non-equilibrium state with a time scale: (a) concentration during drying, when evaporation or a drying front outruns diffusive exchange with the remaining liquid; (b) slow exchange of the layer with the reservoir, for example a layer loaded with Ca^{2+} at one activity and then exposed to another, or Ca^{2+} released by protonation of the sites when the pH changes, which leaves the layer above the bulk chemical potential until diffusion relaxes it; (c) local pH excursions that shift the carbonate speciation at the interface. Each needs transport and kinetics with a stated time scale. They lie outside a thermodynamic screen, which evaluates equilibrium states at registered inputs, and P0 makes no statement about them. What a screen can do is evaluate Ω_{bulk} at each registered state along the drying path, because evaporation does change the bulk activities; that is the bulk driving force at a later state, not a local excess.

7.2 Exact reference: the grand potential of a cap

At fixed T and fixed bulk chemical potentials the open system minimises its grand potential. Forming a cap of radius r on the template transfers $n = V_{\mathrm{cap}}/v_{\mathrm{m}}$ formula units from the reservoir, each at the bulk chemical potentials, creates the cap surface and replaces template–liquid interface by template–nucleus interface over the base:

$$\Delta\Omega_{\mathrm{G}}(r) = -n\,k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}} + \gamma_{\mathrm{nl}}A_{\mathrm{cap}} + (\gamma_{\mathrm{sn}} - \gamma_{\mathrm{sl}})A_{\mathrm{base}} \quad [\mathrm{J}], \tag{32}$$

with $\gamma_{\mathrm{sl}}, \gamma_{\mathrm{sn}}, \gamma_{\mathrm{nl}}$ the grand-canonical (excess grand potential per area) energies at the bulk chemical potentials. Within capillarity this is exact. The reservoir term carries no template dependence, and by Gibbs' construction every effect of the template on the ions around it is an excess per area: there is no other place for it to sit. Equation (32) has the form of Eq. (21), so the factorisation of Section 5 applies with $\Delta\mu = k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}}$ and the in-situ γ_{sl}.

7.3 Gibbs adsorption: where the bound Ca^{2+} goes

Template–liquid interface. At fixed T the Gibbs adsorption equation is $\mathrm{d}\gamma_{\mathrm{sl}} = -\sum_i \Gamma_i\,\mathrm{d}\mu_i$. The Ca^{2+} binding computations use charge-neutral cells, and the adsorbing component is taken to be Ca^{2+} with its compensating charge, that is, a neutral solute. This is an approximation, and Section 4(ii) does not justify it. The cancellation there concerns the ion product: the factors $\mathrm{e}^{\mp 2e\psi/k_{\mathrm{B}}T}$ of the two ions cancel in $a_{\mathrm{Ca}^{2+}}a_{\mathrm{CO}_3^{2-}}$ and in the neutral solid. It says nothing about the double-layer (electrocapillary) part of γ_{sl}. Binding Ca^{2+} to anionic sites changes the surface charge density σ, and with it the free energy of the diffuse layer, which is part of γ_{sl}; for an electrified interface the Gibbs equation carries the additional Lippmann term $-\sigma\,\mathrm{d}\psi_0$, with ψ_0 the surface potential. Equation (33) is the neutral-solute (Gibbs–Langmuir) term only, and the neglected double-layer term is a stated limitation (below). Raising the Ca^{2+} activity from zero (the Ca-free template, $\gamma_{\mathrm{sl}}^{(0)}$) to the regime value at fixed pH and ionic strength, with the Langmuir excess of Eq. (9) and $\mathrm{d}\mu_{\mathrm{Ca}} = RT\,\mathrm{d}\ln c$ (dilute bulk, fixed activity coefficient):

$$\Delta\gamma_{\mathrm{ads}} \equiv \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sl}}^{(0)} = -\int_0^{c_{\mathrm{bulk}}} \Gamma_{\mathrm{Ca}}(c)\,RT\frac{\mathrm{d}c}{c} = -RT\Big[\Gamma_{\mathrm{s}}\ln(1+x) - \delta_{\mathrm{s}}c_{\mathrm{bulk}}\Big] \quad [\mathrm{J\,m^{-2}}], \tag{33}$$

because $\int_0^{c_{\mathrm{bulk}}} \Gamma_{\mathrm{s}}\frac{x(c)}{1+x(c)}\frac{\mathrm{d}c}{c} = \Gamma_{\mathrm{s}}\ln(1+x)$ with $x(c) = \mathcal{E}c/c_{\max}$ linear in c, and $\int_0^{c_{\mathrm{bulk}}} \delta_{\mathrm{s}}c\,\frac{\mathrm{d}c}{c} = \delta_{\mathrm{s}}c_{\mathrm{bulk}}$. Units: $RT\Gamma_{\mathrm{s}}$ is $\mathrm{J\,mol^{-1}\cdot mol\,m^{-2}} = \mathrm{J\,m^{-2}}$, and $\delta_{\mathrm{s}}c_{\mathrm{bulk}}$ is $\mathrm{mol\,m^{-2}}$. In terms of the C2 inputs, $x = \exp(-\Delta G_{\mathrm{seq}}/RT)\,c_{\mathrm{bulk}}\delta_{\mathrm{s}}/\Gamma_{\mathrm{s}}$, so $\Delta\gamma_{\mathrm{ads}}$ is a function of ΔG_{seq}, $a_{\mathrm{Ca}^{2+}}$ (through c_{bulk}), Γ_{s} and δ_{s} only. Its properties:

- $\partial\Delta\gamma_{\mathrm{ads}}/\partial\Delta G_{\mathrm{seq}} = \Gamma_{\mathrm{s}}\Theta > 0$: each site contributes its occupancy, so deeper binding lowers γ_{sl};
- $\partial\Delta\gamma_{\mathrm{ads}}/\partial\mu_{\mathrm{Ca}} = -\Gamma_{\mathrm{Ca}}$: the Gibbs equation itself;
- at strong binding ($x \gg 1$), $\Delta\gamma_{\mathrm{ads}} \to \Gamma_{\mathrm{s}}\Delta G_{\mathrm{seq}} + RT\Gamma_{\mathrm{s}}\ln(c_{\max}/c_{\mathrm{bulk}})$ plus terms that vanish: each site gives its binding free energy, less the free energy of concentrating the ion from the bulk concentration to that of a filled site;
- a template that repels Ca^{2+} ($\Delta G_{\mathrm{seq}} \to +\infty$) has $\Delta\gamma_{\mathrm{ads}} \to +RT\delta_{\mathrm{s}}c_{\mathrm{bulk}} > 0$: negative adsorption raises γ_{sl};
- with no specific interaction ($\Delta G_{\mathrm{seq}} = 0$) $\Delta\gamma_{\mathrm{ads}} = RT\Gamma_{\mathrm{s}}[y - \ln(1+y)]$ with $y = c_{\mathrm{bulk}}\delta_{\mathrm{s}}/\Gamma_{\mathrm{s}}$, a pure site-exclusion correction that vanishes as $y^2/2$ in the dilute limit.

If γ_{sl} is computed directly at the regime activities, the adsorption is already inside it and $\Delta\gamma_{\mathrm{ads}} \equiv 0$, the same rule as for $\Delta\gamma_{\mathrm{w}}$ (Section 6).

Limitation: the double-layer term. Equation (33) omits the change of the diffuse-layer free energy when binding changes the surface charge. In the low-potential (Debye–Hückel) limit of Gouy–Chapman theory, a surface of charge density σ carries a diffuse-layer free energy $\sigma^2/(2\varepsilon_r\varepsilon_0\kappa_D)$ per area, where κ_D^{-1} is the Debye length and $\kappa_D^2 \propto I$, the ionic strength. Units: $C^2\,m^{-4}$ over $F\,m^{-1}{\cdot}m^{-1}$ is $J\,m^{-2}$. At a fixed change of charge the term therefore falls as $I^{-1/2}$: ionic strength screens it, and it vanishes when binding leaves the charge unchanged. P0 does not add it, and does not assign it a sign: it is an unquantified correction to $\Delta\gamma_{ads}$, largest for a template whose Ca^{2+} binding changes its charge strongly at low ionic strength. A γ_{sl} computed in situ with explicit ions contains it, which is one more reason to prefer an in-situ γ_{sl} when one exists.

Template–nucleus interface. γ_{sn} is a grand-canonical excess in the same sense as γ_{sl}: the excess grand potential per area of the template–crystal contact at the regime chemical potentials. Under the contact disc the solution is gone, but the contact need not hold the stoichiometric crystal on one side and the bare template on the other. Ca^{2+} can sit in excess at the contact, for example as an ion bridge between a carboxylate and the crystal surface (carboxylate–Ca–calcite). With the dividing surface at the template and the excess counted relative to the stoichiometric nucleus, such a Ca^{2+} excess $\Gamma_{Ca,sn}$ is inside γ_{sn}, and the Gibbs equation applies to it as it does to γ_{sl}. With $\Gamma_{Ca,sl} \equiv \Gamma_{Ca}$ of Eq. (9) the template–liquid excess,

$$\frac{\partial\gamma_{sn}}{\partial\mu_{Ca}} = -\Gamma_{Ca,sn}, \qquad \frac{\partial(\gamma_{sl}-\gamma_{sn})}{\partial\mu_{Ca}} = -\big(\Gamma_{Ca,sl}-\Gamma_{Ca,sn}\big) \quad [\mathrm{mol\,m^{-2}}]. \tag{34}$$

Units: $J\,m^{-2}$ per $J\,mol^{-1}$ is $mol\,m^{-2}$, an areal excess. The second relation is the result of this subsection. The Young numerator $\gamma_{sl}-\gamma_{sn}$, and with it the sign of the template's effect on the barrier, responds to the Ca^{2+} activity through the difference of the two excesses. Raising $a_{Ca^{2+}}$ raises $\cos\theta$ only if the contact holds more Ca^{2+} excess than the template–liquid interface does. **Both excesses are interfacial; neither is in Ω.**

The screen treats the two excesses differently because its inputs differ. $\Gamma_{Ca,sl}$ is built from C2 (Langmuir, Eq. (9)) and enters through $\Delta\gamma_{ads}$ when γ_{sl} is supplied Ca-free. $\Gamma_{Ca,sn}$ has no C2 counterpart: it is a property of the template–crystal contact, which ΔG_{seq} (transfer from water to a site with water still around it) does not measure. It is carried by γ_{sn} as supplied, so γ_{sn} (or $\gamma_{sn}^{(0)}$ with $\Delta\gamma_w$) is taken from contact data at the regime state, computed or fetched for the template–crystal contact at the regime activities (the C3 inputs). The rule is a rule for the supplied inputs: ΔG_{seq} and $\Delta\gamma_{ads}$ are never added to γ_{sn}. It does not claim $\Gamma_{Ca,sn}=0$. A contact energy supplied at another Ca^{2+} activity would need the shift $-\int\Gamma_{Ca,sn}\,d\mu_{Ca}$, which requires a model of the contact excess that the screen does not have; such an input is flagged as off-state, never corrected with ΔG_{seq}.

Consequence for the barrier (Q1). Young's relation becomes

$$\cos\theta = \frac{\gamma_{sl}^{(0)}+\Delta\gamma_{ads}-\gamma_{sn}^{(0)}-\Delta\gamma_w}{\gamma_{nl}}, \qquad \frac{\partial\cos\theta}{\partial\Delta G_{seq}} = \frac{\Gamma_s\Theta}{\gamma_{nl}} > 0 \quad [1;\ \mathrm{mol\,J^{-1}}]. \tag{35}$$

At fixed $\gamma_{sl}^{(0)}$ and γ_{sn} (the contact energy as supplied at the regime state), stronger binding lowers $\cos\theta$ and raises θ, f and ΔG_{het}^*. Ions held at the template–liquid interface must be stripped from the area the nucleus covers, which is a cost; the physics is that of an adsorbed inhibitor. A Ca-binding template promotes nucleation only if $\gamma_{sl}-\gamma_{sn}$ stays positive after the adsorption term, that is, if γ_{sn} falls by more than γ_{sl} does. By Eq. (34) the sign of the template's response to Ca^{2+} is the sign of $\Gamma_{Ca,sn}-\Gamma_{Ca,sl}$: the Ca^{2+} excess at the contact less that at the template–liquid interface. The first enters the screen through γ_{sn} as supplied, the second through $\Delta\gamma_{ads}$. This is the possibility v1 flagged in Q1, now settled: bookkeeping (L) gave $\Delta G_{seq}<0$ a barrier-lowering effect it does not have (sign-flip test SF2).

7.4 The no-double-counting proof

Statement. In the screen composite

$$\Delta G^{*}_{\mathrm{het}} = f\Big(\theta\big[\gamma^{(0)}_{\mathrm{sl}} + \Delta\gamma_{\mathrm{ads}}(\Delta G_{\mathrm{seq}}, c_{\mathrm{bulk}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}}),\ \gamma^{(0)}_{\mathrm{sn}},\ \Delta\gamma_{\mathrm{w}}\big]\Big)\ \frac{16\pi\gamma^{3}_{\mathrm{nl}} v^{2}_{\mathrm{m}}}{3\big(k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}}\big)^{2}} \quad [\mathrm{J}] \qquad (36)$$

every effect of the template on the barrier is counted exactly once.

Proof. (1) *Exactness.* Equation (32) is the free energy of cap formation in the open system within capillarity; Eq. (36) is its stationary value after the factorisation of Section 5, with $\gamma_{\mathrm{sl}} = \gamma^{(0)}_{\mathrm{sl}} + \Delta\gamma_{\mathrm{ads}}$. (2) *Channel separation.* Ω_{bulk} depends on bulk activities only, so $\partial\Omega/\partial(\text{any template input}) = 0$. All template dependence is in γ_{sl} and γ_{sn}, hence in θ; γ_{nl}, V_{m} and T are template-independent and may appear in both factors. (3) *Single entry of the binding.* ΔG_{seq}, Γ_{s} and δ_{s} enter γ_{sl} through Eq. (33) only, and only when γ_{sl} is built from $\gamma^{(0)}_{\mathrm{sl}}$. They do not enter γ_{sn}, which is supplied from contact data at the regime state with its own excess $\Gamma_{\mathrm{Ca,sn}}$ inside it (template–nucleus interface, above), and they do not enter Ω (Eq. (15)). The water-displacement correction enters γ_{sn} once, by the convention of Section 6. (4) Hence each template-dependent input appears once, in one interfacial energy, and only through θ. This completes the proof.

The checker verifies the syntactic form of (2) and (3) on the contract: the template inputs are partitioned into the θ channel, the Ω channel holds bulk inputs only, and ΔG_{seq}, Γ_{s}, δ_{s} reach the composite only through $\Delta\gamma_{\mathrm{ads}}$.

The two errors, demonstrated. The contract keeps two wrong composites, named and never used for a verdict (C3 item 13):

- `DG_het_local_ideal` (contract v1): Ca-free $\gamma^{(0)}_{\mathrm{sl}}$ with Ω_{li}. The binding is counted once, but in a channel that does not exist at equilibrium, and it lowers the barrier where the exact result raises it.
- `DG_het_double`: the in-situ γ_{sl} (Gibbs term inside) with Ω_{li}. The same binding free energy is charged to γ_{sl} and credited to $\Delta\mu$.

Both exceed the exact driving force by $k_{\mathrm{B}}T\ln(c_{\mathrm{local}}/c_{\mathrm{bulk}})$ per formula unit, which is $-\Delta G_{\mathrm{seq}}/N_{\mathrm{A}}$ in the dilute limit. The barrier of either is smaller than that of the exact Ω-factor by $\big[\ln\Omega_{\mathrm{bulk}}/\ln\Omega_{\mathrm{li}}\big]^{2}$, and `DG_het_double` reduces to the v1 composite without the Gibbs term and to the screen composite without the local credit (both checked). Together with the double application of $\Delta\gamma_{\mathrm{w}}$ of Section 6, these are the double-counting demonstrations.

7.5 Rule for the screen

1. The driving force is $\Delta\mu = k_{\mathrm{B}}T\ln\Omega_{\mathrm{bulk}}$ at the regime activities of Ca^{2+} and CO_3^{2-}. Neither $\mathcal{E}$, c_{local} nor Ω_{li} enters the barrier.
2. C2's ΔG_{seq} maps into the barrier only through $\Delta\gamma_{\mathrm{ads}}(\Delta G_{\mathrm{seq}}, c_{\mathrm{bulk}}, \Gamma_{\mathrm{s}}, \delta_{\mathrm{s}})$ of Eq. (33), added once to the Ca-free $\gamma^{(0)}_{\mathrm{sl}}$. If γ_{sl} is supplied at the regime activities, nothing is added.
3. γ_{sn} is the grand-canonical energy of the template–crystal contact at the regime chemical potentials, taken from contact data at the regime state (plus $\Delta\gamma_{\mathrm{w}}$ if it comes from a reference without the displaced water). Any Ca^{2+} excess at the contact is inside it, $\partial\gamma_{\mathrm{sn}}/\partial\mu_{\mathrm{Ca}} = -\Gamma_{\mathrm{Ca,sn}}$. ΔG_{seq} and $\Delta\gamma_{\mathrm{ads}}$ are never added to γ_{sn}. This is a rule for the supplied input, not a claim that $\Gamma_{\mathrm{Ca,sn}} = 0$.
4. Carbonate needs no correction: a mean potential cancels in Ω and in $\Delta\mu$.

5. The same rule governs R-slurry stability: dissolution of $CaCO_3$ is decided by Ω_{bulk}, the quantity C3 uses (Section 9). An evaluation at an "enriched" state is a non-equilibrium sensitivity, never an equilibrium result.

8 Film conformality

Physics. A barrier must be continuous: one through-path lets electrolyte reach Al. A low nucleation barrier says where nuclei form easily, not whether they spread into a film. Whether the mineral spreads or beads on the template is a wetting balance, and whether beads merge before they grow thick is a competition between how many islands form and how far each spreads.

Spreading coefficient. The free-energy change per unit area when a thick layer of the nucleus phase replaces the template–liquid interface is

$$S_{\mathrm{spread}} = \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} - \gamma_{\mathrm{nl}} = \gamma_{\mathrm{nl}}(\cos\theta - 1) \quad [\mathrm{J\,m^{-2}}], \tag{37}$$

the second form holding on the partial-wetting branch where Young has a solution. Units: a sum of interfacial free energies. Since $\cos\theta \le 1$, a finite contact angle always has $S_{\mathrm{spread}} \le 0$. The sign condition is: $S_{\mathrm{spread}} \ge 0$ means no finite angle exists, $\theta \to 0$, $f \to 0$, and the phase grows layer by layer (Frank–van der Merwe); $S_{\mathrm{spread}} < 0$ means $\theta > 0$ and growth by islands (Volmer–Weber). Island-then-layer growth (Stranski–Krastanov) needs a strain or misfit energy that this isotropic framework does not contain. The registered $\Delta\gamma_{\mathrm{w}}$ enters S_{spread} through γ_{sn}, with the same sign as in θ: a displacement cost makes spreading harder.

Coalescence of islands. With $S_{\mathrm{spread}} < 0$, a film forms only when islands meet. Let the islands be equal caps of contact angle θ on a square lattice of density N_{s} (spacing $\ell = N_{\mathrm{s}}^{-1/2}$). A cap of base radius ρ has height $h = \rho\tan(\theta/2)$. Neighbours touch when $2\rho = \ell$, and the base discs then cover only $\pi/4$ of the area; the square cells are covered when $\rho = \ell/\sqrt{2}$. The two geometric thicknesses are

$$h_{\mathrm{c}} = \frac{\tan(\theta/2)}{2\sqrt{N_{\mathrm{s}}}}, \qquad h_{\mathrm{cl}} = \frac{\tan(\theta/2)}{\sqrt{2N_{\mathrm{s}}}} = \sqrt{2}\,h_{\mathrm{c}} \quad [\mathrm{m}]. \tag{38}$$

Units: $N_{\mathrm{s}}^{-1/2}$ is a length. Both vanish as $\theta \to 0$ and diverge as $\theta \to \pi$. They are geometric estimates that treat caps as growing self-similarly up to contact; they are arguments, not measurements, and N_{s} comes from a nucleation-and-growth model (item 17), not from thermodynamics. The kinetic counterpart, the thickness at which islands actually coalesce in a kinetic Monte Carlo or rate-equation model of nucleation and lateral growth (charter C4, candidate (c)), is a different quantity and carries its own name, h_{coal} (cp2 fix F9). h_{c} keeps its geometric meaning throughout; P0 does not compute h_{coal}.

What a contact angle does and does not imply about coverage. A small θ means thin islands and early closure for a given N_{s}. It does not fix N_{s}, which depends on the nucleation rate relative to lateral growth; it says nothing about places where the template is absent, where the bare-oxide angle applies; and it is silent on porosity, on dissolution and reprecipitation of the amorphous phase, and on cracking when a hydrated film shrinks on drying. A C3 band $f(\theta) \le \frac{1}{2}$ admits θ up to $\pi/2$, which is island growth; conformality is therefore a separate criterion (C4, design flag 2).

The three C4 candidates and what each contains (lane note N1). (a) A small-angle bound $\theta \le \theta_{\max}$ and (b) $S_{\mathrm{spread}} \ge -s$ (with $s \ge 0$ a margin) carry the same information. On the partial-wetting branch Eq. (37) gives $S_{\mathrm{spread}} = \gamma_{\mathrm{nl}}(\cos\theta - 1)$, so

$$S_{\mathrm{spread}} \ge -s \iff \cos\theta \ge 1 - \frac{s}{\gamma_{\mathrm{nl}}} \iff \theta \le \theta_{\max} = \arccos\Big(1 - \frac{s}{\gamma_{\mathrm{nl}}}\Big), \tag{39}$$

because cos is decreasing on $[0, \pi]$ and $\gamma_{\mathrm{nl}} > 0$. Both test one quantity, the Young numerator $\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}}$ relative to γ_{nl}; they differ only in strictness. $S_{\mathrm{spread}} \geq 0$ ($s = 0$, $\theta_{\max} = 0$) is the strict end, complete wetting; it is also the better-posed form, because it is defined when Young has no solution ($\gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} > \gamma_{\mathrm{nl}}$), where θ is not. A finite $\theta_{\max}$ needs a physical scale, and the only one the theory supplies is the coalescence condition: closure below $d_{\max}$ at island density N_{s} requires $\tan(\theta/2) \leq d_{\max}\sqrt{2N_{\mathrm{s}}}$, i.e. $\theta_{\max} = 2\arctan\left(d_{\max}\sqrt{2N_{\mathrm{s}}}\right)$. (c) The coalescence condition, here in its geometric form $h_{\mathrm{cl}} \leq d_{\max}$ (the charter's kinetic form uses h_{coal}), equivalently

$$N_{\mathrm{s}} \geq N_{\mathrm{s,cl}} = \frac{\tan^2(\theta/2)}{2d_{\max}^2} \quad [\mathrm{m}^{-2}], \tag{40}$$

brings in N_{s}, set by nucleation relative to lateral growth, and $d_{\max}$, set by electron transport (Section 10). Neither (a) nor (b) contains either quantity, so (c) is the independent condition. C4 therefore registers one criterion from the (a)/(b) family and uses (c) as the independent supporting condition (`theory/thresholds.md`).

Complete wetting is excluded on a hydrophilic template (cp3). Interface stability gives $\gamma_{\mathrm{sn}} \geq 0$: a negative interfacial free energy between two immiscible solids would make the system create contact without limit. Equation (37) then bounds the spreading coefficient by the two in-water energies alone,

$$S_{\mathrm{spread}} = \gamma_{\mathrm{sl}} - \gamma_{\mathrm{sn}} - \gamma_{\mathrm{nl}} \leq \gamma_{\mathrm{sl}} - \gamma_{\mathrm{nl}} \quad [\mathrm{J\,m^{-2}}], \tag{41}$$

so complete wetting, $S_{\mathrm{spread}} \geq 0$, needs $\gamma_{\mathrm{sl}} \geq \gamma_{\mathrm{nl}}$: the template–water interface must cost at least as much per area as the mineral–water interface. A hydrophilic template is one that water wets at least as favourably as it wets the mineral, $\gamma_{\mathrm{sl}} \leq \gamma_{\mathrm{nl}}$; for it $S_{\mathrm{spread}} \leq -\gamma_{\mathrm{sn}} < 0$ for every admissible $\gamma_{\mathrm{sn}} > 0$ (equality only on the boundary $\gamma_{\mathrm{sn}} = 0$, $\gamma_{\mathrm{sl}} = \gamma_{\mathrm{nl}}$), whatever the template–nucleus contact is. Complete wetting is excluded, and a continuous film on such a template can form only by coalescence of islands, condition (40). The mechanism's own premise pushes the same way: Ca^{2+} binding lowers γ_{sl} ($\Delta\gamma_{\mathrm{ads}} < 0$ for a binding template, Section 7). The screen reports $P(\gamma_{\mathrm{sl}} > \gamma_{\mathrm{nl}})$, the largest mass the spreading route can carry, and C4 is feasible through either route (`theory/thresholds.md`).

9 Thermodynamic stability

Physics. A layer that is thermodynamically unstable in its own environment dissolves or reacts. The environment differs between regimes, so stability is two questions: in the slurry, whether $CaCO_3$ and the native oxide survive alkaline water at the regime pH and potential; in the cell, whether the phases survive the Li reservoir at the collector's potential. Both are screens: necessary, not sufficient, and blind to morphology and kinetics.

R-slurry: aqueous Pourbaix. Each species ϕ (solid or dissolved) is brought to a common reference by exchanging O and H with water, protons and electrons at the regime pH and potential. Per non-O, non-H atom,

$$\Psi_\phi = \frac{1}{n_{\mathrm{M}}}\Big[G_\phi - n_{\mathrm{O}}\mu_{\mathrm{H_2O}} - RT\ln 10\,(2n_{\mathrm{O}} - n_{\mathrm{H}})\,\mathrm{pH} \\ - (2n_{\mathrm{O}} - n_{\mathrm{H}} + Q)FE_{\mathrm{SHE}}\Big] \quad [\mathrm{J\,mol^{-1}}], \qquad \Delta G_{\mathrm{pbx}} = \Psi_\phi - \Psi_{\mathrm{hull}} \geq 0. \tag{42}$$

Units: G_ϕ and $\mu_{\mathrm{H_2O}}$ are $\mathrm{J\,mol^{-1}}$, $RT\ln 10\,\mathrm{pH}$ is $\mathrm{J\,mol^{-1}}$, and FE_{SHE} is $\mathrm{C\,mol^{-1}{\cdot}V} = \mathrm{J\,mol^{-1}}$. Species with $2n_{\mathrm{O}} - n_{\mathrm{H}} + Q > 0$ are favoured by raising E_{SHE}, and species with $2n_{\mathrm{O}} - n_{\mathrm{H}} > 0$ by raising pH. Ψ_{hull} is the lower envelope over all species and mixtures of the same metal composition, and $\Delta G_{\mathrm{pbx}} = 0$ means stable. *What it can say:* whether the phase is the equilibrium state at the chosen (pH, E_{SHE}) and dissolved-ion activities. *What it cannot say:* rates, passivation, local pH at the interface, morphology.

Dissolution is decided by Ω_{bulk}, the quantity C3 uses. Carbon is held in the C(IV) carbonate system: the regime fixes $a_{\mathrm{CO_3^{2-}}}$, and reduction of dissolved carbonate to CH_4 is not observed in aqueous processing, so reduced-carbon phases are not admitted (an assumption; the construction with carbon redox allowed is a labelled sensitivity). At fixed C(IV), dissolving a $CaCO_3$ layer is the reaction $CaCO_3(s) \rightarrow Ca^{2+} + CO_3^{2-}$ at the regime activities, with free energy $RT \ln K_{\mathrm{sp}} - RT \ln(a_{\mathrm{Ca^{2+}}} a_{\mathrm{CO_3^{2-}}}) = -RT \ln \Omega_{\mathrm{bulk}}$ per mole of $CaCO_3$, where K_{sp} and the activities are those of Eq. (10) (activities from the speciation of Section 4). The layer dissolves when $\Omega_{\mathrm{bulk}} < 1$ and is stable against dissolution when $\Omega_{\mathrm{bulk}} \geq 1$. This is the same Ω_{bulk} that drives C3: a slurry state that cannot nucleate $CaCO_3$ is a state in which a $CaCO_3$ layer dissolves, and by Section 4 enrichment at the template changes neither. Written as a hull distance per non-O, non-H atom ($n_{\mathrm{M}} = 2$ for $CaCO_3$), with the dissolved ions on the hull,

$$\Delta G_{\mathrm{pbx}}\big|_{\mathrm{diss}} = \frac{RT}{n_{\mathrm{M}}} \ln \frac{1}{\Omega_{\mathrm{bulk}}} \quad (\Omega_{\mathrm{bulk}} < 1) \quad [\mathrm{J\,mol^{-1}}]. \tag{43}$$

Two consequences follow, and they are why a computed hull distance cannot decide dissolution. First, a hull tolerance $\tau = n_{\mathrm{at}} e_{\mathrm{f}} / n_{\mathrm{M}}$ reports dissolution only below $\ln \Omega_{\mathrm{bulk}} = -n_{\mathrm{at}} e_{\mathrm{f}} / (RT)$: since RT is small against a typical formation-energy error times n_{at}, the tolerance hides several decades of undersaturation. Second, an error δ in the computed dissolution free energy per mole shifts the apparent K_{sp} by $\delta/(RT \ln 10)$ decades, so the database's dissolved-ion reference adds a bias of its own. Dissolution is therefore decided from Ω_{bulk} with the measured calcite K_{sp} (the registered framing's K_{sp} for ACC is reported beside it as a sensitivity), and the screen must show that this test reports dissolution where the solution is undersaturated (C5 item 21, Q8).

Competing phases and the oxide baseline. The hull of Eq. (42), with C(IV) fixed and the dissolved Ca-bearing species removed (dissolution is the Ω_{bulk} test), answers the remaining question: whether another solid lies below $CaCO_3$. It carries the tolerance of the computed formation energies. The amorphous framing lies above the calcite hull by its offset, so ACC is never on the hull. Adding the offset to the edge makes the ACC test equal to the calcite-level test: $\Psi_{\mathrm{ACC}} = \Psi_{\mathrm{calcite}} + \Delta_{\mathrm{ACC}}/n_{\mathrm{M}}$ and the hull is unchanged, so the test asks whether $CaCO_3$ is stable as a phase at the regime activities, up to the framing's metastability. Whether ACC itself dissolves is Ω with the ACC solubility product, a labelled sensitivity (cp2 fix F6). The native oxide's Pourbaix state at the same fixed C(IV) is the bare-oxide baseline and does not gate.

R-cell: Li-open grand potential. In the cell the solvent is not water and the reference is Li^+/Li. Holding the collector at E_{Li} fixes the Li chemical potential,

$$\mu_{\mathrm{Li}} = \mu^{\circ}_{\mathrm{Li}} - F E_{\mathrm{Li}} \quad [\mathrm{J\,mol^{-1}}], \qquad \Phi_{\phi} = G_{\phi} - n_{\mathrm{Li}} \mu_{\mathrm{Li}} \quad [\mathrm{J\,mol^{-1}}]. \tag{44}$$

For a reaction $\phi \rightarrow \sum_k \nu_k \psi_k + \Delta n\,\mathrm{Li}$ that releases Δn Li to the reservoir, Li conservation gives $\Delta\Phi(E_{\mathrm{Li}}) = \sum_k \nu_k \Phi_k - \Phi_{\phi} = \Delta G_{\mathrm{r}} - \Delta n\, F E_{\mathrm{Li}}$, with ΔG_{r} the reaction free energy against Li metal. The phase is stable against that reaction while $\Delta\Phi > 0$, so each reaction sets one limit,

$$E^* = \frac{\Delta G_{\mathrm{r}}}{\Delta n\, F} \quad [\mathrm{V}], \qquad \begin{cases} \Delta n > 0 \text{ (delithiation, oxidation)}: & E_{\mathrm{Li}} < E^* \;\rightarrow\; E_{\mathrm{ox}} = \min E^*, \\ \Delta n < 0 \text{ (lithiation, reduction)}: & E_{\mathrm{Li}} > E^* \;\rightarrow\; E_{\mathrm{red}} = \max E^*. \end{cases} \tag{45}$$

Units: $\mathrm{J\,mol^{-1}}$ over $\mathrm{C\,mol^{-1}}$ is V. *What it can say:* whether a phase reacts with Li at the collector's potential. *What it cannot say:* oxidation that exchanges no Li. A Li-free phase such as $CaCO_3$ has no reaction with $\Delta n > 0$, so its Li-open E_{ox} is unbounded: the construction declares it stable at any anodic potential by construction. Anodic stability of $CaCO_3$ and of the Al phases needs oxidation reactions that release CO_2, O_2 or dissolved Ca^{2+} and Al^{3+} into the electrolyte, which require the electrolyte species as additional open components (Q9). Kinetics and electrolyte oxidation are outside both screens.

The oxidation sink is fixed by the electrolyte (Q9). When the collector oxidises $CaCO_3$, the carbonate gives up its electrons and leaves as CO_2 and O_2. The Ca^{2+} must go somewhere, and the cell offers only what the electrolyte contains: Li^+, the salt anion A^- and the solvent. Electroneutrality pairs each released Ca^{2+} with two A^-, either as the solid salt CaA_2 or as solvated ions, and the electrons reach the Li^+/Li reference as Li plated from the salt LiA. Referenced to Li^+/Li, the reactions are

$$CaCO_3 + 2\,LiA \rightarrow CaA_2 + CO_2 + \tfrac{1}{2}O_2 + 2\,Li, \qquad Al_2O_3 + 6\,LiA \rightarrow 2\,AlA_3 + \tfrac{3}{2}O_2 + 6\,Li, \quad (46)$$

each with $\Delta n > 0$ and a limit $E^* = \Delta G_r/(\Delta n\, F)$ from Eq. (45). The sink is therefore not a free choice: it is the anion of the registered electrolyte (`charter/regimes.yaml`, `electrolyte_composition`). The rule for choosing among candidate salts, once `lit/params.yaml` lists the fluorine-free salts: (1) only the anions of the registered salt or salts are admissible; (2) the oxidation limit is the minimum over the admissible sinks and over the products each can form, because a phase is unstable if any accessible reaction is downhill; (3) the true product is the lower in free energy of the solid salt and the solvated ions, so a limit computed with the solid salt alone is an upper bound on the true onset: a falsification computed with it stands, and a pass computed with it is inconclusive, so that a pass counts only with the solvated sink; (4) a sink whose thermochemistry is missing makes the limit *undetermined*, which never counts as a pass; (5) where the CO_2 (or O_2) reference is uncertain, the limit is evaluated under each registered reference scheme: a pass must hold under the scheme with the lowest onset, a falsification under every scheme, and anything else is inconclusive (cp2 fix F5); since cp3 this rule applies only where the all-experimental route is missing (next paragraph). *Limitation:* the electrolyte's own anodic limit is a separate quantity. Above it the solvent and salt oxidise first, and it is not assessed here, so a pass says nothing about the stability of the electrolyte.

Which thermochemistry decides the limit (cp3). $E^* = \Delta G_r/(\Delta n\, F)$ is only as good as ΔG_r. A computed reaction energy that combines database solids with a gas reference carries the error of the carbonate formation energy and of the CO_2 reference, and the two reference schemes bracket the measured reaction energies from opposite sides. Where every reactant and product is tabulated, the all-experimental route has no such error and governs; its margin m_{expt} is the experimental uncertainty budget (the tabulated uncertainties per transferred electron, plus a bound on any neglected solid entropy, such as that of a sink salt tabulated by its enthalpy only, taken from analog salts). A computed scheme corrected by its measured benchmark error reproduces that same experimental route, so it is a check of the bookkeeping and not an independent confirmation. Because the solid-salt sink is an upper bound on the onset, the governing rule is: falsified if $E_{\mathrm{ox}}^{\mathrm{expt}} + m_{\mathrm{expt}} < E_{\mathrm{cell}}^{\mathrm{max}}$ with either sink; feasible only if $E_{\mathrm{ox}}^{\mathrm{expt}} - m_{\mathrm{expt}} \geq E_{\mathrm{cell}}^{\mathrm{max}}$ with the solvated sink. The charge cutoff $E_{\mathrm{cell}}^{\mathrm{max}}$ is a property of the cathode, so the verdict is a function of it, with crossovers at $E_{\mathrm{ox}}^{\mathrm{expt}} \pm m_{\mathrm{expt}}$. *Why the carbonate-oxidation reading is physical.* For the analogous reaction of Li_2CO_3, $Li_2CO_3 \rightarrow 2Li + CO_2 + \frac{1}{2}O_2$ referenced to Li^+/Li, the measured oxidation onset lies near the experimental equilibrium limit of the same reaction (W1 key `li2co3_oxidation_onset` against the experimental route of W5's gas-reference benchmark): a solid carbonate in a non-aqueous cell oxidises close to its thermodynamic limit, with little kinetic overpotential. The window limit of $CaCO_3$ is therefore read as the onset of an observable reaction, not merely as a bound that kinetics could postpone indefinitely.

10 The insulating-film limitation

Physics. $CaCO_3$ is an electronic insulator. The film lies between the cathode coating and the Al collector, and every electron of the cathode current must cross it. A film that is continuous but too thick blocks the collector, so conformality cannot be bought with thickness. Two transport limits bound the admissible thickness; the numbers come later, through macros.

Ohmic limit. If electrons cross by bulk conduction with resistivity ρ_{e}, the area-specific resistance and the bound set by an admissible $R_{A,\max}$ are

$$R_A = \rho_{\mathrm{e}} d \quad [\Omega\,\mathrm{m}^2], \qquad d \le d_{\max}^{\mathrm{ohm}} = \frac{R_{A,\max}}{\rho_{\mathrm{e}}} \quad [\mathrm{m}]. \tag{47}$$

Units: $\Omega\,\mathrm{m}$ times m is $\Omega\,\mathrm{m}^2$; $\Omega\,\mathrm{m}^2$ over $\Omega\,\mathrm{m}$ is m.

Tunnelling limit. For a thin film the current is carried by direct tunnelling through a barrier of height ϕ_{B}. In the WKB approximation for a rectangular barrier the transmission decays as $\mathrm{e}^{-2\kappa d}$, so

$$j = j_0\,\mathrm{e}^{-2\kappa d} \quad [\mathrm{A\,m^{-2}}], \qquad \kappa = \frac{\sqrt{2 m_{\mathrm{e}} \phi_{\mathrm{B}}}}{\hbar} \quad [\mathrm{m}^{-1}], \qquad d \le d_{\max}^{\mathrm{tun}} = \frac{\ln(j_0/j_{\mathrm{req}})}{2\kappa} \quad [\mathrm{m}]. \tag{48}$$

Units: $\sqrt{\mathrm{kg}\cdot\mathrm{J}}/(\mathrm{J\,s}) = \mathrm{m}^{-1}$; the logarithm is of a ratio of current densities. The bound is positive only for $j_0 > j_{\mathrm{req}}$. The prefactor j_0, the effective mass and the barrier height are inputs with large uncertainty (Q10), and defects or hopping can only raise the current, so Eq. (48) is a conservative bound for a defect-free film.

Combined with conformality. The film must close before it exceeds the admissible thickness:

$$h_{\mathrm{cl}} = \frac{\tan(\theta/2)}{\sqrt{2N_{\mathrm{s}}}} \le d_{\max} = \max\left(d_{\max}^{\mathrm{ohm}},\, d_{\max}^{\mathrm{tun}}\right) \quad [\mathrm{m}]. \tag{49}$$

This is a stated limitation (C4 item 18), not a performance figure. It contains no statement about cell resistance, rate or capacity.

11 Summary: links and the inputs they consume

Physics. Each link reduces to a small set of inputs. Listing them once is what lets the parameter table (W1) fetch every input and the implementation (W4) consume every one, with nothing typed and nothing missing. Table 3 is that list; the ids are those of `theory/symbols.yaml`, and the functions are those of `theory/contract.yaml`.

Link	Regime	Contract functions	Inputs consumed (symbol ids)	Source
C1 adhesion	R-slurry	W_adh_wet, W_adh_wet_from_pmf, W_adh_dry_fn	Delta_F_pmf, A_c; comparator gamma_o, gamma_p, gamma_op, gamma_w	computed PMF; W1 for comparators
C2 sequestration	R-slurry	Delta_G_seq_cycle, E_enr, c_max, c_local, x_bind, Theta_occ, Gamma_Ca, Delta_gamma_ads	Delta_G_site, pKa, pH, T, c_bulk, Gamma_s, delta_s, V_site	computed PMF; W1 for pKa, pH, c_bulk
Super-saturation	R-slurry	Omega_bulk, Delta_mu; demonstration: Omega_local_ideal, Delta_mu_local_ideal	a_Ca, a_CO3, K_sp, T	W1 (activities from speciation at the regime pH and ionic strength)
C3 nucleation	R-slurry	r_star, DG_hom, f_theta, DG_het, cos_theta_young, cos_theta_template, gamma_sl_insitu, theta_young, B_acc, N_nuc; audit: gamma_eff_deck, cos_theta_deck	gamma_nl, gamma_sl0 (or gamma_sl in situ), gamma_sn0, Delta_gamma_w, V_m, T, J0_het, t_proc, N_req, with Ω_{bulk} and Delta_gamma_ads	W1 priors and computed slabs; both framings registered (cp3)
C4 conformality	R-slurry	S_spread, h_c, h_cl, N_s_closure	C3 energies, N_s, d_max	nucleation-and-growth model (item 17); N_nuc as an upper estimate
C4 film bound	R-cell	R_A, d_max_ohm, kappa, d_max_tun	rho_e, R_A_max, phi_B, m_e, j_0, j_req, d_film	W1; R_A_max, j_req registered at G2
C5 Pourbaix	R-slurry	Psi_pbx, Delta_G_pbx	G_phase (with the ACC offset), mu_H2O, N_O, N_H, N_M, Q_charge, pH, E_SHE, T	W1 thermochemistry; materials database
C5 window	R-cell	mu_Li, Phi_Li, E_window	G_phase, N_Li, mu_Li0, E_Li, Delta_G_r, Delta_n_Li	materials database

Table 3: Each link, the contract functions that implement it, and the inputs it consumes. Every input is fetched or computed; none is a result of the study.